11 GRAPHS



Figure 11.1 Cyclists speed toward the finish line. (credit: ewan traveler, Flickr)

Chapter Outline

11.1 Use the Rectangular Coordinate System
11.2 Graphing Linear Equations
11.3 Graphing with Intercepts
11.4 Understand Slope of a Line

Graphs

Which cyclist will win the race? What will the winning time be? How many seconds will separate the winner from the runner-up? One way to summarize the information from the race is by creating a graph. In this chapter, we will discuss the basic concepts of graphing. The applications of graphing go far beyond races. They are used to present information in almost every field, including healthcare, business, and entertainment.

11.1 Use the Rectangular Coordinate System

Learning Objectives

By the end of this section, you will be able to:

- Plot points on a rectangular coordinate system
- Identify points on a graph
- Verify solutions to an equation in two variables
- Complete a table of solutions to a linear equation
- Find solutions to linear equations in two variables

BE PREPARED 11.1 Before you get started, take this readiness quiz.

Evaluate: $x + 3$ when $x = -1$.
If you missed this problem, review Example 3.23.

BE PREPARED 11.2 Evaluate: $2x - 5y$ when $x = 3, y = -2$.
If you missed this problem, review Example 3.56.

BE PREPARED 11.3 Solve for y: $40 - 4y = 20$.
If you missed this problem, review Example 8.20.

Plot Points on a Rectangular Coordinate System

Many maps, such as the Campus Map shown in Figure 11.2, use a grid system to identify locations. Do you see the numbers $1, 2, 3$, and 4 across the top and bottom of the map and the letters A, B, C, and D along the sides? Every location on the map can be identified by a number and a letter.

For example, the Student Center is in section 2B. It is located in the grid section above the number 2 and next to the letter B. In which grid section is the Stadium? The Stadium is in section 4D.

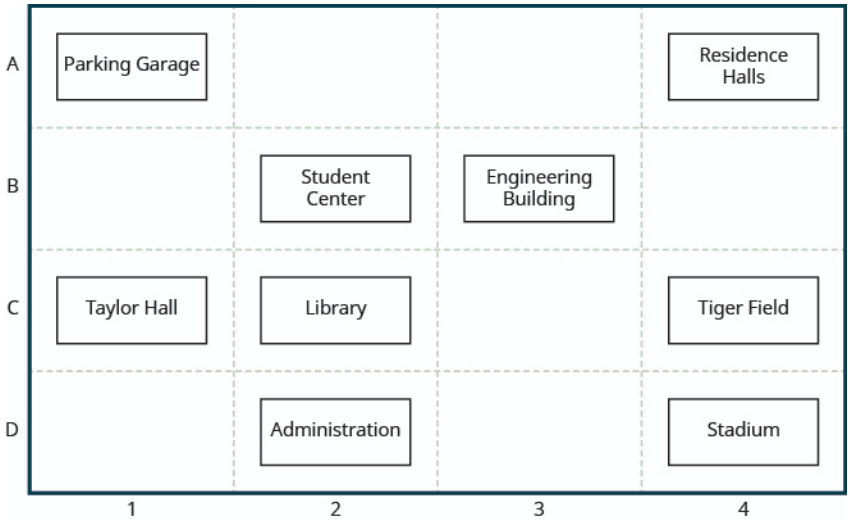


Figure 11.2

EXAMPLE 11.1

Use the map in Figure 11.2.

ⓐ Find the grid section of the Residence Halls. ⓑ What is located in grid section 4C?

Solution

ⓐ Read the number below the Residence Halls, 4, and the letter to the side, A. So the Residence Halls are in grid section 4A.

ⓑ Find 4 across the bottom of the map and C along the side. Look below the 4 and next to the C. Tiger Field is in grid section 4C.

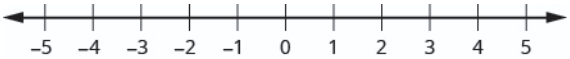
TRY IT 11.1 Use the map in Figure 11.2.

ⓐ Find the grid section of Taylor Hall. ⓑ What is located in section 3B?

TRY IT 11.2 Use the map in Figure 11.2.

ⓐ Find the grid section of the Parking Garage. ⓑ What is located in section 2C?

Just as maps use a grid system to identify locations, a grid system is used in algebra to show a relationship between two variables in a rectangular coordinate system. To create a rectangular coordinate system, start with a horizontal number line. Show both positive and negative numbers as you did before, using a convenient scale unit. This horizontal number line is called the ***x*-axis**.



Now, make a vertical number line passing through the x-axis at 0. Put the positive numbers above 0 and the negative numbers below 0. See Figure 11.3. This vertical line is called the ***y*-axis**.

Vertical grid lines pass through the integers marked on the x-axis. Horizontal grid lines pass through the integers marked on the y-axis. The resulting grid is the rectangular coordinate system.

The rectangular coordinate system is also called the x-y plane, the coordinate plane, or the Cartesian coordinate system

(since it was developed by a mathematician named René Descartes.)

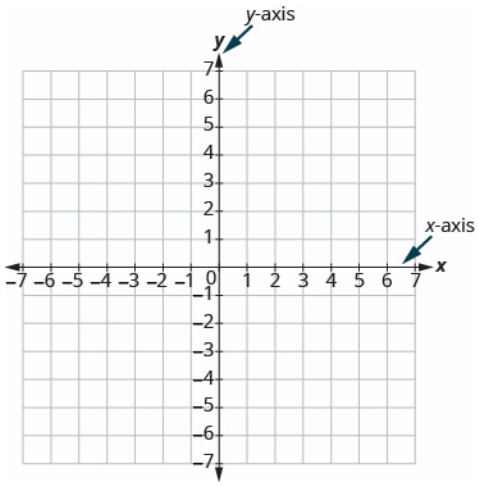


Figure 11.3 The rectangular coordinate system.

The x-axis and the y-axis form the rectangular coordinate system. These axes divide a plane into four areas, called **quadrants**. The quadrants are identified by Roman numerals, beginning on the upper right and proceeding counterclockwise. See Figure 11.4.

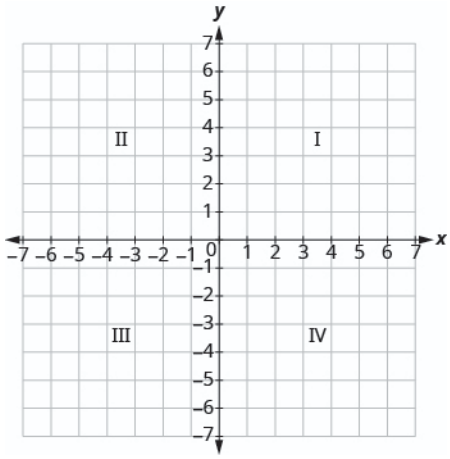


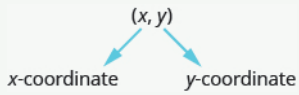
Figure 11.4 The four quadrants of the rectangular coordinate system

In the rectangular coordinate system, every point is represented by an **ordered pair**. The first number in the ordered pair is the *x*-coordinate of the point, and the second number is the *y*-coordinate of the point.

Ordered Pair

An ordered pair, (x, y) gives the coordinates of a point in a rectangular coordinate system.

The first number is the x-coordinate.
The second number is the y-coordinate.



So how do the coordinates of a point help you locate a point on the x-y plane?

Let's try locating the point $(2, 5)$. In this ordered pair, the x-coordinate is 2 and the y-coordinate is 5.

We start by locating the x value, 2, on the x-axis. Then we lightly sketch a vertical line through $x = 2$, as shown in Figure 11.5.

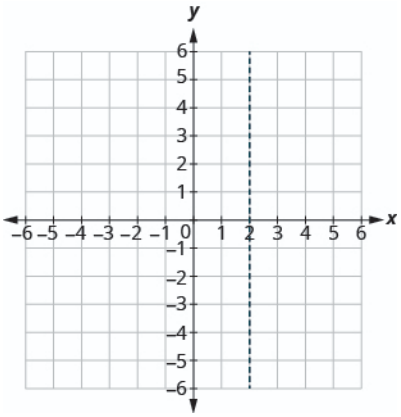


Figure 11.5

Now we locate the y value, 5, on the y-axis and sketch a horizontal line through $y = 5$. The point where these two lines meet is the point with coordinates $(2, 5)$. We plot the point there, as shown in Figure 11.6.

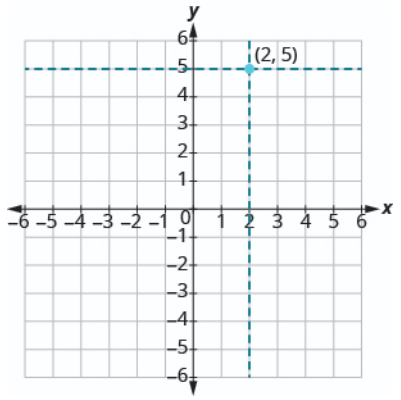


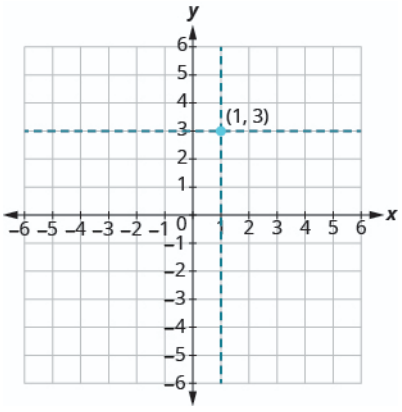
Figure 11.6

EXAMPLE 11.2

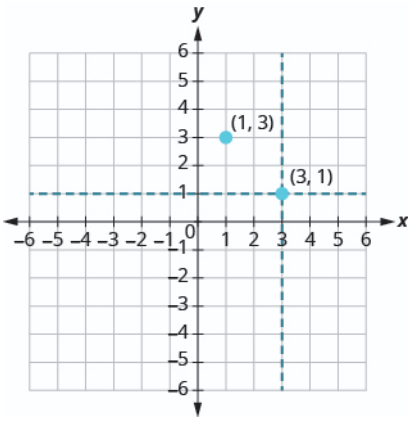
Plot $(1, 3)$ and $(3, 1)$ in the same rectangular coordinate system.

Solution

The coordinate values are the same for both points, but the x and y values are reversed. Let's begin with point $(1, 3)$. The x-coordinate is 1 so find 1 on the x-axis and sketch a vertical line through $x = 1$. The y-coordinate is 3 so we find 3 on the y-axis and sketch a horizontal line through $y = 3$. Where the two lines meet, we plot the point $(1, 3)$.



To plot the point $(3, 1)$, we start by locating 3 on the x-axis and sketch a vertical line through $x = 3$. Then we find 1 on the y-axis and sketch a horizontal line through $y = 1$. Where the two lines meet, we plot the point $(3, 1)$.



Notice that the order of the coordinates does matter, so, $(1, 3)$ is not the same point as $(3, 1)$.

TRY IT 11.3 Plot each point on the same rectangular coordinate system: $(5, 2)$, $(2, 5)$.

TRY IT 11.4 Plot each point on the same rectangular coordinate system: $(4, 2)$, $(2, 4)$.

EXAMPLE 11.3

Plot each point in the rectangular coordinate system and identify the quadrant in which the point is located:

ⓐ $(-1, 3)$ ⓑ $(-3, -4)$ ⓒ $(2, -3)$ ⓓ $\left(3, \frac{5}{2}\right)$

Solution

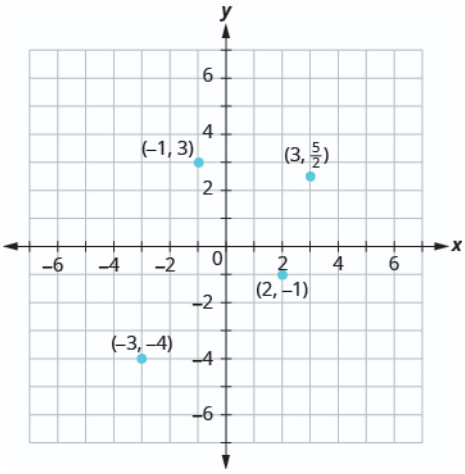
The first number of the coordinate pair is the x-coordinate, and the second number is the y-coordinate.

ⓐ Since $x = -1, y = 3$, the point $(-1, 3)$ is in Quadrant II.

ⓑ Since $x = -3, y = -4$, the point $(-3, -4)$ is in Quadrant III.

ⓒ Since $x = 2, y = -1$, the point $(2, -1)$ is in Quadrant IV.

ⓓ Since $x = 3, y = \frac{5}{2}$, the point $\left(3, \frac{5}{2}\right)$ is in Quadrant I. It may be helpful to write $\frac{5}{2}$ as the mixed number, $2\frac{1}{2}$, or decimal, 2.5. Then we know that the point is halfway between 2 and 3 on the y-axis.



TRY IT 11.5 Plot each point on a rectangular coordinate system and identify the quadrant in which the point is located.

ⓐ $(-2, 1)$ ⓑ $(-3, -1)$ ⓒ $(4, -4)$ ⓓ $\left(-4, \frac{3}{2}\right)$

TRY IT 11.6 Plot each point on a rectangular coordinate system and identify the quadrant in which the point is located.

ⓐ $(-4, 1)$ ⓑ $(-2, 3)$ ⓒ $(2, -5)$ ⓓ $\left(-3, \frac{5}{2}\right)$

How do the signs affect the location of the points?

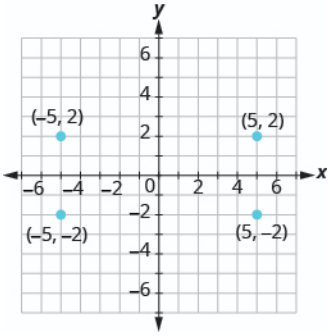
EXAMPLE 11.4

Plot each point:

ⓐ $(-5, 2)$ ⓑ $(-5, -2)$ ⓒ $(5, 2)$ ⓓ $(5, -2)$

Solution

As we locate the x-coordinate and the y-coordinate, we must be careful with the signs.



TRY IT 11.7 Plot each point:

ⓐ $(4, -3)$ ⓑ $(4, 3)$ ⓒ $(-4, -3)$ ⓓ $(-4, 3)$

TRY IT 11.8 Plot each point:

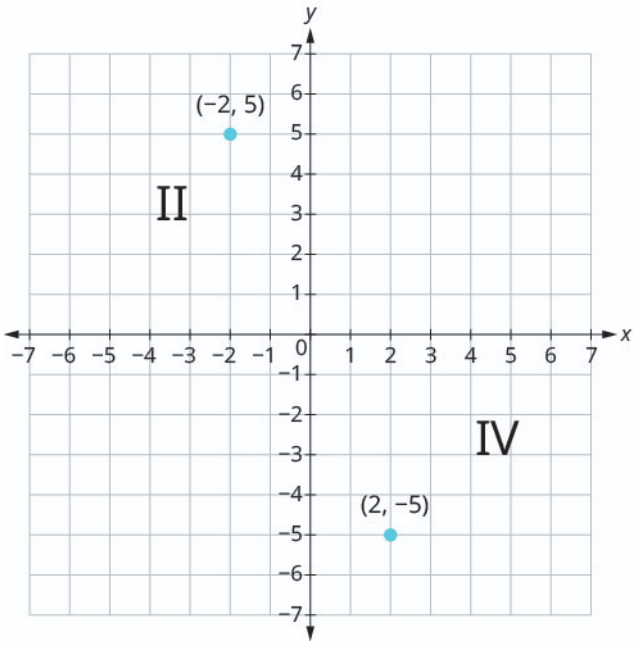
ⓐ $(-1, 4)$ ⓑ $(1, 4)$ ⓒ $(1, -4)$ ⓓ $(-1, -4)$

You may have noticed some patterns as you graphed the points in the two previous examples.

For each point in Quadrant IV, what do you notice about the signs of the coordinates?

What about the signs of the coordinates of the points in the third quadrant? The second quadrant? The first quadrant?

Can you tell just by looking at the coordinates in which quadrant the point (−2, 5) is located? In which quadrant is (2, −5) located?



We can summarize sign patterns of the quadrants as follows. Also see Figure 11.7.

Quadrant I	Quadrant II	Quadrant III	Quadrant IV
(x,y)	(x,y)	(x,y)	(x,y)
(+,+)	(−,+)	(−,−)	(+,−)

Table 11.1

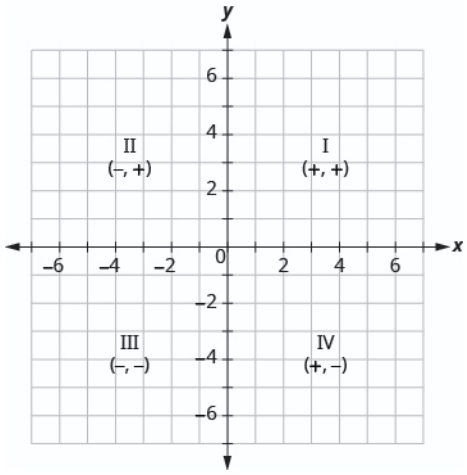


Figure 11.7

What if one coordinate is zero? Where is the point $(0, 4)$ located? Where is the point $(-2, 0)$ located? The point $(0, 4)$ is on the *y*-axis and the point $(-2, 0)$ is on the *x*-axis.

Points on the Axes

Points with a y-coordinate equal to 0 are on the x-axis, and have coordinates $(a, 0)$.

Points with an x-coordinate equal to 0 are on the y-axis, and have coordinates $(0, b)$.

What is the ordered pair of the point where the axes cross? At that point both coordinates are zero, so its ordered pair

is $(0, 0)$. The point has a special name. It is called the *origin*.

The Origin

The point $(0, 0)$ is called the **origin**. It is the point where the *x*-axis and *y*-axis intersect.

EXAMPLE 11.5

Plot each point on a coordinate grid:

ⓐ $(0, 5)$ ⓑ $(4, 0)$ ⓒ $(-3, 0)$ ⓓ $(0, 0)$ ⓔ $(0, -1)$

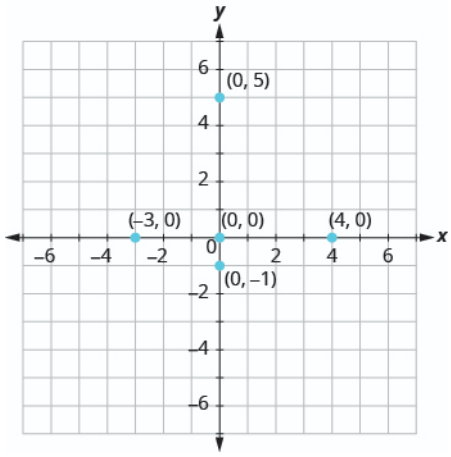
Solution

ⓐ Since $x = 0$, the point whose coordinates are $(0, 5)$ is on the y-axis.

ⓑ Since $y = 0$, the point whose coordinates are $(4, 0)$ is on the x-axis.

ⓒ Since $y = 0$, the point whose coordinates are $(-3, 0)$ is on the x-axis.

ⓓ Since $x = 0$ and $y = 0$, the point whose coordinates are $(0, 0)$ is the origin.



ⓔ Since $x = 0$, the point whose coordinates are $(0, -1)$ is on the y-axis.

TRY IT 11.9 Plot each point on a coordinate grid:

ⓐ $(4, 0)$ ⓑ $(-2, 0)$ ⓒ $(0, 0)$ ⓓ $(0, 2)$ ⓔ $(0, -3)$

TRY IT 11.10 Plot each point on a coordinate grid:

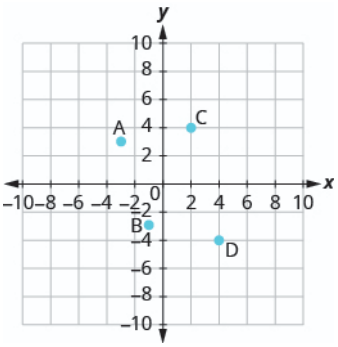
ⓐ $(-5, 0)$ ⓑ $(3, 0)$ ⓒ $(0, 0)$ ⓓ $(0, -1)$ ⓔ $(0, 4)$

Identify Points on a Graph

In algebra, being able to identify the coordinates of a point shown on a graph is just as important as being able to plot points. To identify the *x*-coordinate of a point on a graph, read the number on the *x*-axis directly above or below the point. To identify the *y*-coordinate of a point, read the number on the *y*-axis directly to the left or right of the point. Remember, to write the ordered pair using the correct order (x, y).

EXAMPLE 11.6

Name the ordered pair of each point shown:



Solution

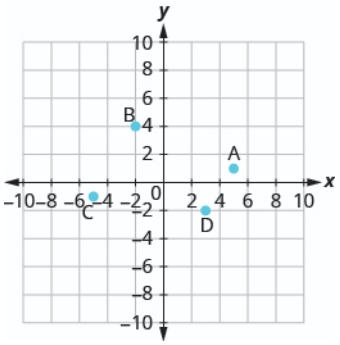
Point A is above -3 on the x-axis, so the x-coordinate of the point is -3. The point is to the left of 3 on the y-axis, so the y-coordinate of the point is 3. The coordinates of the point are $(-3, 3)$.

Point B is below -1 on the x-axis, so the x-coordinate of the point is -1. The point is to the left of -3 on the y-axis, so the y-coordinate of the point is -3. The coordinates of the point are $(-1, -3)$.

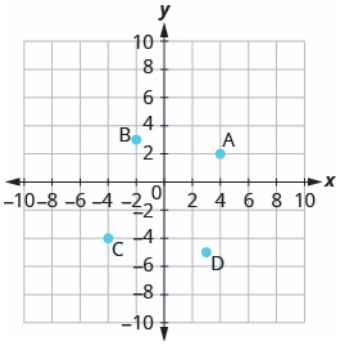
Point C is above 2 on the x-axis, so the x-coordinate of the point is 2. The point is to the right of 4 on the y-axis, so the y-coordinate of the point is 4. The coordinates of the point are $(2, 4)$.

Point D is below 4 on the $x-$ axis, so the x-coordinate of the point is 4. The point is to the right of -4 on the y-axis, so the y-coordinate of the point is -4. The coordinates of the point are $(4, -4)$.

TRY IT 11.11 Name the ordered pair of each point shown:

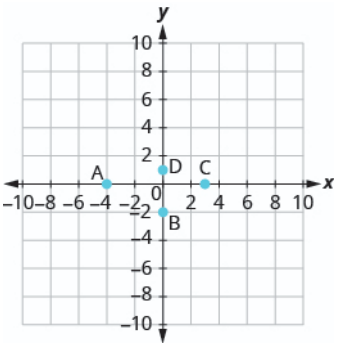


TRY IT 11.12 Name the ordered pair of each point shown:



EXAMPLE 11.7

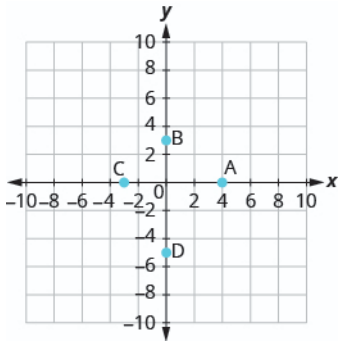
Name the ordered pair of each point shown:



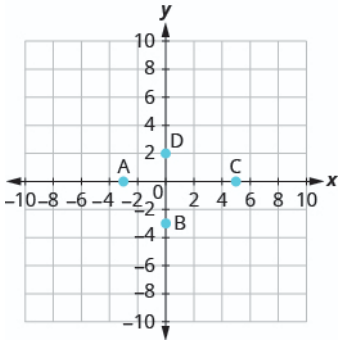
Solution

Point A is on the *x*-axis at $x = -4$.	The coordinates of point A are $(-4, 0)$.
Point B is on the *y*-axis at $y = -2$	The coordinates of point B are $(0, -2)$.
Point C is on the *x*-axis at $x = 3$.	The coordinates of point C are $(3, 0)$.
Point D is on the *y*-axis at $y = 1$.	The coordinates of point D are $(0, 1)$.

TRY IT 11.13 Name the ordered pair of each point shown:



TRY IT 11.14 Name the ordered pair of each point shown:



Verify Solutions to an Equation in Two Variables

All the equations we solved so far have been equations with one variable. In almost every case, when we solved the equation we got exactly one solution. The process of solving an equation ended with a statement such as $x = 4$. Then we checked the solution by substituting back into the equation.

Here's an example of a linear equation in one variable, and its one solution.

$$3x + 5 = 17$$

$$3x = 12$$
$$x = 4$$

But equations can have more than one variable. Equations with two variables can be written in the general form $Ax + By = C$. An equation of this form is called a linear equation in two variables.

Linear Equation

An equation of the form $Ax + By = C$, where A and B are not both zero, is called a linear equation in two variables.

Notice that the word "line" is in linear.

Here is an example of a linear equation in two variables, x and y:

$$Ax + By = C$$
$$x + 4y = 8$$

$A = 1$, $B = 4$, $C = 8$

Is $y = -5x + 1$ a linear equation? It does not appear to be in the form $Ax + By = C$. But we could rewrite it in this form.

	$y = -5x + 1$
Add $5x$ to both sides.	$y + 5x = -5x + 1 + 5x$
Simplify.	$y + 5x = 1$
Use the Commutative Property to put it in $Ax + By = C$.	$Ax + By = C$ $5x + y = 1$

By rewriting $y = -5x + 1$ as $5x + y = 1$, we can see that it is a linear equation in two variables because it can be written in the form $Ax + By = C$.

Linear equations in two variables have infinitely many solutions. For every number that is substituted for x, there is a corresponding y value. This pair of values is a **solution to the linear equation** and is represented by the ordered pair (x, y). When we substitute these values of x and y into the equation, the result is a true statement because the value on the left side is equal to the value on the right side.

Solution to a Linear Equation in Two Variables

An ordered pair (x, y) is a solution to the linear equation $Ax + By = C$, if the equation is a true statement when the x- and y-values of the ordered pair are substituted into the equation.

EXAMPLE 11.8

Determine which ordered pairs are solutions of the equation $x + 4y = 8$:

ⓐ $(0, 2)$ ⓑ $(2, -4)$ ⓒ $(-4, 3)$

✓ **Solution**

Substitute the x- and y-values from each ordered pair into the equation and determine if the result is a true statement.

(a) $(0, 2)$	(b) $(2, -4)$	(c) $(-4, 3)$
$x = 0, y = 2$	$x = 2, y = -4$	$x = -4, y = 3$
$x + 4y = 8$	$x + 4y = 8$	$x + 4y = 8$
$0 + 4 \cdot 2 \stackrel{?}{=} 8$	$2 + 4(-4) \stackrel{?}{=} 8$	$-4 + 4 \cdot 3 \stackrel{?}{=} 8$
$0 + 8 \stackrel{?}{=} 8$	$2 + (-16) \stackrel{?}{=} 8$	$-4 + 12 \stackrel{?}{=} 8$
$8 = 8$ ✓	$-14 \neq 8$	$8 = 8$ ✓
$(0, 2)$ is a solution.	$(2, -4)$ is not a solution.	$(-4, 3)$ is a solution.

TRY IT 11.15 Determine which ordered pairs are solutions to the given equation: $2x + 3y = 6$

(a) $(3, 0)$ (b) $(2, 0)$ (c) $(6, -2)$

TRY IT 11.16 Determine which ordered pairs are solutions to the given equation: $4x - y = 8$

(a) $(0, 8)$ (b) $(2, 0)$ (c) $(1, -4)$

EXAMPLE 11.9

Determine which ordered pairs are solutions of the equation. $y = 5x - 1$:

(a) $(0, -1)$ (b) $(1, 4)$ (c) $(-2, -7)$

Solution

Substitute the x- and y-values from each ordered pair into the equation and determine if it results in a true statement.

(a) $(0, -1)$	(b) $(1, 4)$	(c) $(-2, -7)$
$x = 0, y = -1$	$x = 1, y = 4$	$x = -2, y = -7$
$y = 5x - 1$	$y = 5x - 1$	$y = 5x - 1$
$-1 \stackrel{?}{=} 5(0) - 1$	$4 \stackrel{?}{=} 5(1) - 1$	$-7 \stackrel{?}{=} 5(-2) - 1$
$-1 \stackrel{?}{=} 0 - 1$	$4 \stackrel{?}{=} 5 - 1$	$-7 \stackrel{?}{=} -10 - 1$
$-1 = -1$ ✓	$4 = 4$ ✓	$-7 \neq -11$
$(0, -1)$ is a solution.	$(1, 4)$ is a solution.	$(-2, -7)$ is not a solution.

TRY IT 11.17 Determine which ordered pairs are solutions of the given equation: $y = 4x - 3$

(a) $(0, 3)$ (b) $(1, 1)$ (c) $(1, 0)$

TRY IT 11.18 Determine which ordered pairs are solutions of the given equation: $y = -2x + 6$

(a) $(0, 6)$ (b) $(1, 4)$ (c) $(-2, -2)$

Complete a Table of Solutions to a Linear Equation

In the previous examples, we substituted the x- and y-values of a given ordered pair to determine whether or not it was a solution to a linear equation. But how do we find the ordered pairs if they are not given? One way is to choose a value for x and then solve the equation for y. Or, choose a value for y and then solve for x.

We'll start by looking at the solutions to the equation $y = 5x - 1$ we found in Example 11.9. We can summarize this information in a table of solutions.

$y = 5x - 1$		
x	y	(x, y)
0	-1	$(0, -1)$
1	4	$(1, 4)$

To find a third solution, we'll let $x = 2$ and solve for y.

	$y = 5x - 1$
Substitute $x = 2$.	$y = 5(2) - 1$
Multiply.	$y = 10 - 1$
Simplify.	$y = 9$

The ordered pair is a solution to $y = 5x - 1$. We will add it to the table.

$y = 5x - 1$		
x	y	(x, y)
0	-1	$(0, -1)$
1	4	$(1, 4)$
2	9	$(2, 9)$

We can find more solutions to the equation by substituting any value of x or any value of y and solving the resulting equation to get another ordered pair that is a solution. There are an infinite number of solutions for this equation.

EXAMPLE 11.10

Complete the table to find three solutions to the equation $y = 4x - 2$:

$y = 4x - 2$		
x	y	(x, y)
0		
-1		
2		

Solution

Substitute $x = 0, x = -1$, and $x = 2$ into $y = 4x - 2$.

$x = 0$	$x = -1$	$x = 2$
$y = 4x - 2$	$y = 4x - 2$	$y = 4x - 2$
$y = 4 \cdot 0 - 2$	$y = 4(-1) - 2$	$y = 4 \cdot 2 - 2$
$y = 0 - 2$	$y = -4 - 2$	$y = 8 - 2$
$y = -2$	$y = -6$	$y = 6$
$(0, -2)$	$(-1, -6)$	$(2, 6)$

The results are summarized in the table.

$y = 4x - 2$		
x	y	(x, y)
0	-2	$(0, -2)$
-1	-6	$(-1, -6)$
2	6	$(2, 6)$

TRY IT 11.19 Complete the table to find three solutions to the equation: $y = 3x - 1$.

$y = 3x - 1$		
x	y	(x, y)
0		
-1		
2		

TRY IT 11.20 Complete the table to find three solutions to the equation: $y = 6x + 1$

$y = 6x + 1$		
x	y	(x, y)
0		
1		
-2		

EXAMPLE 11.11

Complete the table to find three solutions to the equation $5x - 4y = 20$:

$5x - 4y = 20$		
x	y	(x, y)
0		
	0	
	5	

Solution

$x = 0$	$y = 0$	$y = 5$
$5x - 4y = 20$	$5x - 4y = 20$	$5x - 4y = 20$
$5 \cdot 0 - 4y = 20$	$5x - 4 \cdot 0 = 20$	$5x - 4 \cdot 5 = 20$
$0 - 4y = 20$	$5x - 0 = 20$	$5x - 20 = 20$
$-4y = 20$	$5x = 20$	$5x = 40$
$y = -5$	$x = 4$	$x = 8$
$(0, -5)$	$(4, 0)$	$(8, 5)$

The results are summarized in the table.

$5x - 4y = 20$		
x	y	(x, y)
0	-5	$(0, -5)$
4	0	$(4, 0)$
8	5	$(8, 5)$

TRY IT 11.21 Complete the table to find three solutions to the equation: $2x - 5y = 20$.

$2x-5y=20$		
x	y	(x,y)
0		
	0	
-5		

TRY IT 11.22 Complete the table to find three solutions to the equation: $3x-4y=12$.

$3x-4y=12$		
x	y	(x,y)
0		
	0	
-4		

Find Solutions to Linear Equations in Two Variables

To find a solution to a linear equation, we can choose any number we want to substitute into the equation for either x or y. We could choose $1, 100, 1{,}000$, or any other value we want. But it's a good idea to choose a number that's easy to work with. We'll usually choose 0 as one of our values.

EXAMPLE 11.12

Find a solution to the equation $3x+2y=6$.

Solution

Step 1: Choose any value for one of the variables in the equation.		We can substitute any value we want for x or any value for y. Let's pick $x=0$. What is the value of y if $x=0$?
Step 2: Substitute that value into the equation. Solve for the other variable.	Substitute 0 for x. Simplify. Divide both sides by 2.	$3x+2y=6$ $3\cdot 0+2y=6$ $0+2y=6$ $2y=6$ $y=3$

Step 3: Write the solution as an ordered pair.	So, when $x = 0, y = 3$.	This solution is represented by the ordered pair $(0, 3)$.
Step 4: Check.	Substitute $x = 0, y = 3$ into the equation $3x + 2y = 6$. Is the result a true equation? Yes!	$3x + 2y = 6$ $3 \cdot 0 + 2 \cdot 3 \stackrel{?}{=} 6$ $0 + 6 \stackrel{?}{=} 6$ $6 = 6$ ✓

TRY IT 11.23 Find a solution to the equation: $4x + 3y = 12$.

TRY IT 11.24 Find a solution to the equation: $2x + 4y = 8$.

We said that linear equations in two variables have infinitely many solutions, and we've just found one of them. Let's find some other solutions to the equation $3x + 2y = 6$.

EXAMPLE 11.13

Find three more solutions to the equation $3x + 2y = 6$.

Solution

To find solutions to $3x + 2y = 6$, choose a value for x or y. Remember, we can choose any value we want for x or y. Here we chose 1 for x, and 0 and -3 for y.

Substitute it into the equation.	$y = 0$ $3x + 2y = 6$ $3x + 2(0) = 6$	$x = 1$ $3x + 2y = 6$ $3(1) + 2y = 6$	$y = -3$ $3x + 2y = 6$ $3x + 2(-3) = 6$
Simplify. Solve.	$3x + 0 = 6$ $3x = 6$	$3 + 2y = 6$ $2y = 3$	$3x - 6 = 6$ $3x = 12$
	$x = 2$	$y = \frac{3}{2}$	$x = 4$
Write the ordered pair.	$(2, 0)$	$(1, \frac{3}{2})$	$(4, -3)$

Check your answers.

$(2, 0)$	$(1, \frac{3}{2})$	$(4, -3)$
$3x + 2y = 6$	$3x + 2y = 6$	$3x + 2y = 6$
$3 \cdot 2 + 2 \cdot 0 \stackrel{?}{=} 6$	$3 \cdot 1 + 2 \cdot \frac{3}{2} \stackrel{?}{=} 6$	$3 \cdot 4 + 2(-3) \stackrel{?}{=} 6$
$6 + 0 \stackrel{?}{=} 6$	$3 + 3 \stackrel{?}{=} 6$	$12 + (-6) \stackrel{?}{=} 6$
$6 = 6$ ✓	$6 = 6$ ✓	$6 = 6$ ✓

So $(2, 0)$, $\left(1, \frac{3}{2}\right)$ and $(4, -3)$ are all solutions to the equation $3x + 2y = 6$. In the previous example, we found that $(0, 3)$ is a solution, too. We can list these solutions in a table.

$3x + 2y = 6$		
x	y	(x, y)
0	3	$(0, 3)$
2	0	$(2, 0)$
1	$\frac{3}{2}$	$(1, \frac{3}{2})$
4	-3	$(4, -3)$

TRY IT 11.25 Find three solutions to the equation: $2x + 3y = 6$.

TRY IT 11.26 Find three solutions to the equation: $4x + 2y = 8$.

Let's find some solutions to another equation now.

EXAMPLE 11.14

Find three solutions to the equation $x - 4y = 8$.

Solution

	$x - 4y = 8$	$x - 4y = 8$	$x - 4y = 8$
Choose a value for x or y.	$x = 0$	$y = 0$	$y = 3$
Substitute it into the equation.	$0 - 4y = 8$	$x - 4 \cdot 0 = 8$	$x - 4 \cdot 3 = 8$
Solve.	$-4y = 8$ $y = -2$	$x - 0 = 8$ $x = 8$	$x - 12 = 8$ $x = 20$
Write the ordered pair.	$(0, -2)$	$(8, 0)$	$(20, 3)$

So $(0, -2)$, $(8, 0)$, and $(20, 3)$ are three solutions to the equation $x - 4y = 8$.

$x - 4y = 8$		
x	y	(x, y)
0	-2	$(0, -2)$
8	0	$(8, 0)$
20	3	$(20, 3)$

Remember, there are an infinite number of solutions to each linear equation. Any point you find is a solution if it makes the equation true.

TRY IT 11.27 Find three solutions to the equation: $4x + y = 8$.

TRY IT 11.28 Find three solutions to the equation: $x + 5y = 10$.

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Plotting Points (https://openstaxcollege.org/l/24plotpoints)

Identifying Quadrants (https://openstaxcollege.org/l/24quadrants)

Verifying Solution to Linear Equation (https://openstaxcollege.org/l/24verlineq)

SECTION 11.1 EXERCISES

Practice Makes Perfect

Plot Points on a Rectangular Coordinate System

In the following exercises, plot each point on a coordinate grid.

1. $(3, 2)$

2. $(4, 1)$

3. $(1, 5)$

4. $(3, 4)$

5. $(4, 1), (1, 4)$

6. $(3, 2), (2, 3)$

7. $(3, 4), (4, 3)$

In the following exercises, plot each point on a coordinate grid and identify the quadrant in which the point is located.

8. ⓐ $(-4, 2)$ ⓑ $(-1, -2)$ ⓒ $(3, -5)$ ⓓ $\left(2, \frac{5}{2}\right)$

9. ⓐ $(-2, -3)$ ⓑ $(3, -3)$ ⓒ $(-4, 1)$ ⓓ $\left(1, \frac{3}{2}\right)$

10. ⓐ $(-1, 1)$ ⓑ $(-2, -1)$ ⓒ $(1, -4)$ ⓓ $\left(3, \frac{7}{2}\right)$

In the following exercises, plot each point on a coordinate grid.

11. ⓐ $(3, -2)$ ⓑ $(-3, 2)$ ⓒ $(-3, -2)$ ⓓ $(3, 2)$

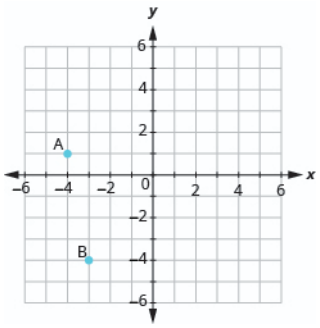
12. ⓐ $(4, -1)$ ⓑ $(-4, 1)$ ⓒ $(-4, -1)$ ⓓ $(4, 1)$

13. 1. ⓐ $(-2, 0)$
2. ⓑ $(-3, 0)$
3. ⓒ $(0, 4)$
4. ⓓ $(0, 2)$

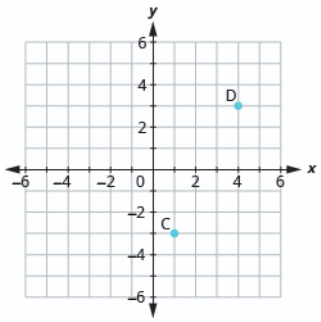
Identify Points on a Graph

In the following exercises, name the ordered pair of each point shown.

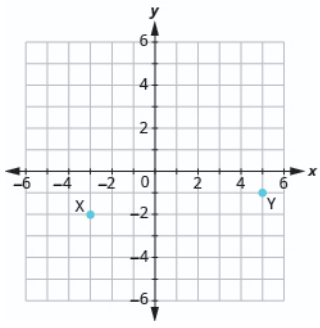
14.



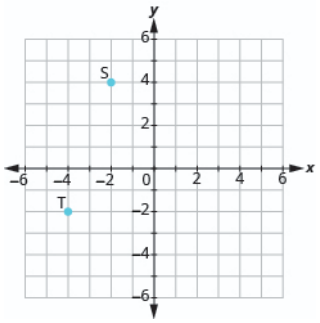
15.



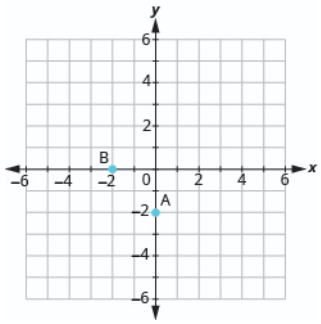
16.



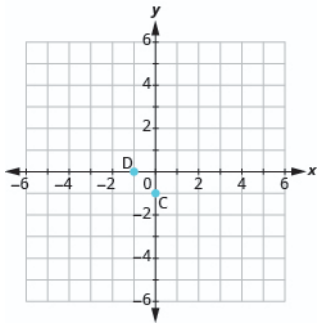
17.



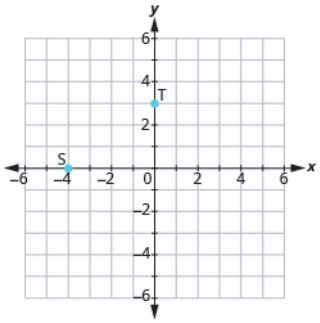
18.



19.



20.



Verify Solutions to an Equation in Two Variables

In the following exercises, determine which ordered pairs are solutions to the given equation.

21. $2x + y = 6$
ⓐ $(1, 4)$ ⓑ $(3, 0)$
ⓒ $(2, 3)$

22. $x + 3y = 9$
ⓐ $(0, 3)$ ⓑ $(6, 1)$
ⓒ $(-3, -3)$

23. $4x - 2y = 8$
ⓐ $(3, 2)$ ⓑ $(1, 4)$
ⓒ $(0, -4)$

24. $3x - 2y = 12$
ⓐ $(4, 0)$ ⓑ $(2, -3)$
ⓒ $(1, 6)$

25. $y = 4x + 3$
ⓐ $(4, 3)$ ⓑ $(-1, -1)$
ⓒ $(\frac{1}{2}, 5)$

26. $y = 2x - 5$
ⓐ $(0, -5)$ ⓑ $(2, 1)$
ⓒ $(\frac{1}{2}, -4)$

27. $y = \frac{1}{2}x - 1$
ⓐ $(2, 0)$ ⓑ $(-6, -4)$
ⓒ $(-4, -1)$

28. $y = \frac{1}{3}x + 1$
ⓐ $(-3, 0)$ ⓑ $(9, 4)$
ⓒ $(-6, -1)$

Find Solutions to Linear Equations in Two Variables

In the following exercises, complete the table to find solutions to each linear equation.

29. $y = 2x - 4$

x	y	(x, y)
-1		
0		
2		

30. $y = 3x - 1$

x	y	(x, y)
-1		
0		
2		

31. $y = -x + 5$

x	y	(x, y)
-2		
0		
3		

32. $y = \frac{1}{3}x + 1$

x	y	(x, y)
0		
3		
6		

33. $y = -\frac{3}{2}x - 2$

x	y	(x, y)
-2		
0		
2		

34. $x + 2y = 8$

x	y	(x, y)
0		
4		
	0	

Everyday Math

35. **Weight of a baby** Mackenzie recorded her baby's weight every two months. The baby's age, in months, and weight, in pounds, are listed in the table, and shown as an ordered pair in the third column.

ⓐ Plot the points on a coordinate grid.

Age	Weight	(x, y)
0	7	$(0, 7)$
2	11	$(2, 11)$
4	15	$(4, 15)$
6	16	$(6, 16)$
8	19	$(8, 19)$
10	20	$(10, 20)$
12	21	$(12, 21)$

ⓑ Why is only Quadrant I needed?

36. **Weight of a child** Latresha recorded her son's height and weight every year. His height, in inches, and weight, in pounds, are listed in the table, and shown as an ordered pair in the third column.

ⓐ Plot the points on a coordinate grid.

Height x	Weight y	(x, y)
28	22	$(28, 22)$
31	27	$(31, 27)$
33	33	$(33, 33)$
37	35	$(37, 35)$
40	41	$(40, 41)$
42	45	$(42, 45)$

ⓑ Why is only Quadrant I needed?

Writing Exercises

37. Have you ever used a map with a rectangular coordinate system? Describe the map and how you used it.

38. How do you determine if an ordered pair is a solution to a given equation?